

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -5

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Recap of Previous Lecture



doubt ?

Topic

Virtual Memory

Topic

Effective Memory Access Time



Topics to be Covered



Topic

Page Replacement Policy

Topic

Optimal, LRU, FIFO Policy





Topic : Question

[GATE-2020]



- #Q. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is 154.5 to 155.5



Topic : Page Replacement Policies

1. First In First Out (FIFO)
2. Optimal Policy
3. Least Recently Used (LRU)
4. Last in First Out (LIFO)
5. Most Recently Used (MRU)
6. Least Frequently Used (LFU)
7. Most Frequently Used (MFU)



Topic : First In First Out (FIFO)

The page which comes first in mm, will be replaced first.

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	4	4	4	5			5	5	
	2	2	2	1	1	1			3	3	
		3	3	3	2	2			2	4	
✓	✓	✓	✓	✓	✓	✓	×	×	✓	✓	×

no. of page faults = 9

$$\text{Page fault rate} = \frac{9}{12} \quad \Bigg| \quad \text{Page hit rate} = \frac{3}{12}$$



Topic : First In First Out (FIFO)

Assume:

- Number of frames = 4 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1			5	5	5	5	4	4
	2	2	2			2	1	1	1	1	5
		3	3			3	3	2	2	2	2
			4			4	4	4	3	3	3
✓	✓	✓	✓	×	×	✓	✓	✓	✓	✓	✓

No. of page faults = 10



Topic : Belady's Anomaly

for some page reference sequences increasing the no. of frames increases number of page faults also.

⇒ FIFO suffers from belady's anomaly.



Topic : First In First Out (FIFO)

Advantages:

1. Simple and easy to implement.
2. Low overhead. *→ To search which page to replace.*

Disadvantages:

1. Poor performance. *→ more page faults*
2. Doesn't consider the frequency of use or last used time, simply replaces the oldest page.
3. Suffers from Belady's Anomaly



Topic : Optimal Policy

replace the page which will not be used for longest period of time.

Assume:

Tie breaker = FIFO

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1			1			3	3	
	2	2	2			2			2	4	
		3	4			5			5	5	
✓	✓	✓	✓	×	×	✓	×	×	✓	✓	×

No. of page faults = 7

Ques) no. of frames = 4 (all empty initially)

optimal

Page reference seq: 4, 2, 4, 6, 3, 4, 9, 2, 9, 2, 8, 6, 2, 4

✓	✓	×	✓	✓	×	✓	×	×	×	✓	×	×	×
4	2	4	6	3	4	9	2	9	2	8	6	2	4
4	4		4	4		4				4			
	2		2	2		2				2			
			6	6		6				6			
				3		9				8			

no. of page faults = 6



Topic : Optimal Policy

Advantages:

1. Easy to Implement
2. Simple data structures are used
3. Highly efficient as it gives minimum number of page faults
4. Doesn't suffer from Belady's Anomaly

Disadvantages:

1. Requires future knowledge of the program $\Rightarrow$ hence not practical
2. Time-consuming to find which page to replace.



Topic : Least Recently Used (LRU)

replace a page which has not been used for longest period of time

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	4	4	4	5			3	3	3
	2	2	2	1	1	1			1	4	4
		3	3	3	2	2			2	2	5
✓	✓	✓	✓	✓	✓	✓	x	x	✓	✓	✓

no. of page faults = 10



Topic : Least Recently Used (LRU)

Advantages:

1. Efficient
2. Doesn't suffer from Belady's Anomaly

Disadvantages:

1. Complex Implementation
2. Expensive
3. Requires hardware support



Topic : Question

- Number of frames = 4
- Pure demand paging used
- Page reference sequence: 5, 7, 0, 1, 7, 6, 7, 2, 1, 6, 7, 6, 1
- Number of page faults for FIFO, optimal and LRU policies?

FIFO

5	5	5	5		6		6		6		
	7	7	7		7		2		2		
		0	0		0		0		7		
			1		1		1		1		
✓	✓	✓	✓	x	✓		✓		✓		



Topic : Question



5, 7, 0, 1, 7, 6, 7, 2, 1, 6, 7, 6, 1

optimal

5	5	5	5	6		6						
	7	7	7	7		7						
		0	0	0		2						
			1	1		1						

LRU

			5	6		6						
			7	7		7						
			0	0		2						
			1	1		1						



Topic : Question

Consider the following page references:

~~2~~, ~~3~~, ~~4~~, ~~5~~, 6, ~~4~~, ~~5~~, ~~3~~, 7, 8, 9, ~~8~~, ~~9~~, ~~8~~, ~~9~~, ~~3~~, ~~7~~, ~~8~~, ~~3~~, 5

Using optimal policy and 4 frames. Memory access time is 2ms without page fault and 40ms with page fault. The effective memory access time for servicing the above page requests is 19.1 ms?

$$\text{Page fault rate} = \frac{9}{20}$$

2	6	6	8	8		8													
3	3	3	3	3		5													
4	4	7	7	7		7													
5	5	5	5	9		9													

$$EMAT = \frac{11}{20} * 2 + \frac{9}{20} * 40 = \frac{22 + 360}{20} = 19.1 \text{ ms}$$



Topic : Last In First Out (LIFO)

replace the page which comes last in mem

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

✓	✓	✓	✓	×	×	✓	×	×	✓	✓	✓
1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1			1			1	1	1
	2	2	2			2			2	2	2
		3	4			5			3	4	5

no. of faults = 8



Topic : Most Recently Used (MRU)

↓
replace the page which has been used most recently.

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

✓	✓	✓	✓	×	×	✓	×	✓	✓	×	×
1	2	3	4	1	2	5	1	2	3	4	5
1	1	1	1			1		2	3		
	2	2	2			5		5	5		
		3	4			4		4	4		

no. of page faults = 7



Topic : Counting Algorithms

- Counting algorithms look at the number of occurrences of a particular page and use this as the criterion for replacement.
- Such counting algorithms includes:
 - LFU (Least Frequently Used)
 - MFU (Most Frequently Used)



Topic : Least Frequently Used (LFU)

Tie breaker $\Rightarrow$ FIFO

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1 2 0 3 0 4 2 3 0 3 2

✓ 1	✓ 2	✓ 0	✓ 3	✗ 0	✓ 4	✓ 2	✓ 3	✗ 0	✗ 3	✗ 2
1	1	1	3		3	2	2			
	2	2	2		4	4	3			
		0	0		0	0	0			

no. of faults = 7

Page	freq
0	1 2 3
1	1
2	1 2 3
3	1 2 3
4	1



Topic : Most Frequently Used (MFU)

Tie breaker = FIFO

Assume:

- Number of frames = 3 (All empty initially)
- Page reference sequence: 1 2 0 3 0 4 2 3 0 3 2

✓1	✓2	✓0	✓3	✗0	✓4	✗2	✗3	✓0	✗3	✓2
1	1	1	3		3			3		2
	2	2	2		2			0		0
		0	0		4			4		4

no of faults = 7

Page	freq
0	✗ 2 3
1	1
2	✗ 2 3
3	✗ 2 3
4	1



Topic : Question

#Q. A virtual memory system has only 2-page frames which are empty initially. Using demand paging the following sequence of page reference is passed through this system.

~~9~~, ~~8~~, ~~7~~, ~~8~~, ~~7~~, 9, 7, 9, 8, 9

Minimum possible number of page faults ? = 5

Optimal

✓ 9	✓ 9	✓ 7	×	7	8
	8	8		9	9



Topic : Question

H.W.

[GATE-2016]



- #Q. Consider a computer system with ten physical page frames. The system is provided with an access sequence $a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20}$ where each a_i is a number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____



Topic : Question

[GATE-2007]



#Q. The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 bytes per page is
0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0410.

Suppose that the memory can store only one page and if x is the address which causes a page fault then the bytes from addresses x to $x + 99$ are loaded on to the memory.

How many page faults will occur?

~~0100 - 0199~~
~~0200 - 0299~~
~~0430 - 0529~~
~~0530 - 0629~~
~~0120 - 0219~~
~~0220 - 0319~~
0320 - 0419

A 0

B 4

C 7 ✓

D 8



Topic : Making Page Reference Sequence

How to get Page no. from given L.A.

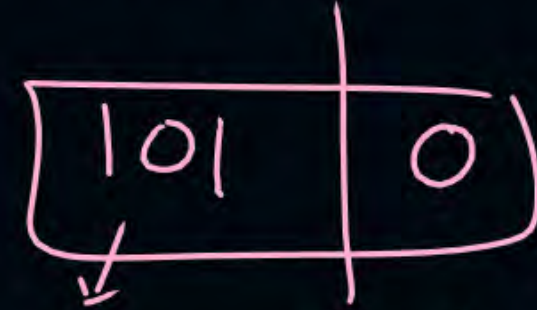
1. If LA given in Binary

2. ——— in hexadecimal



convert

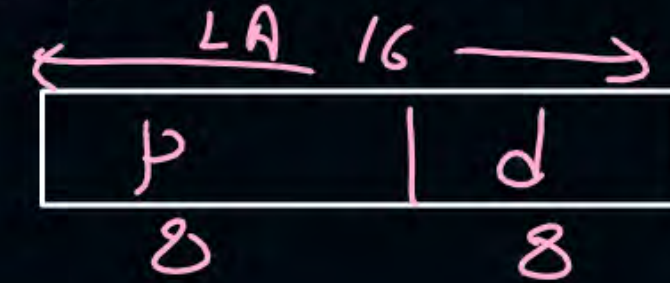
1010



Page no. = $(101)_2 = (5)_{10}$

H.W.

#Q. A demand paged virtual memory system uses 16 bit virtual address, page size of 256 bytes, and has 1Kbyte of main memory. LRU page replacement is implemented using the list, whose current status (page number is decimal) is



For each hexadecimal address in the address sequence given below,

00FF, 010D, 10FF, 11B0
indicate ⁰ ¹ ¹⁶ ¹⁷

1. the new status of the list
2. page faults, if any, and
3. page replacements, if any.

$$00000000 = (0)_{10}$$

$$(00000001)_2 = (1)_{10}$$

$$(00010000)_2 = (16)_{10}$$

$$(00010001)_2 = (17)_{10}$$



2 mins Summary

Topic

Page Replacement Policy

Topic

Optimal, LRU, FIFO Policy





Happy Learning

THANK - YOU

